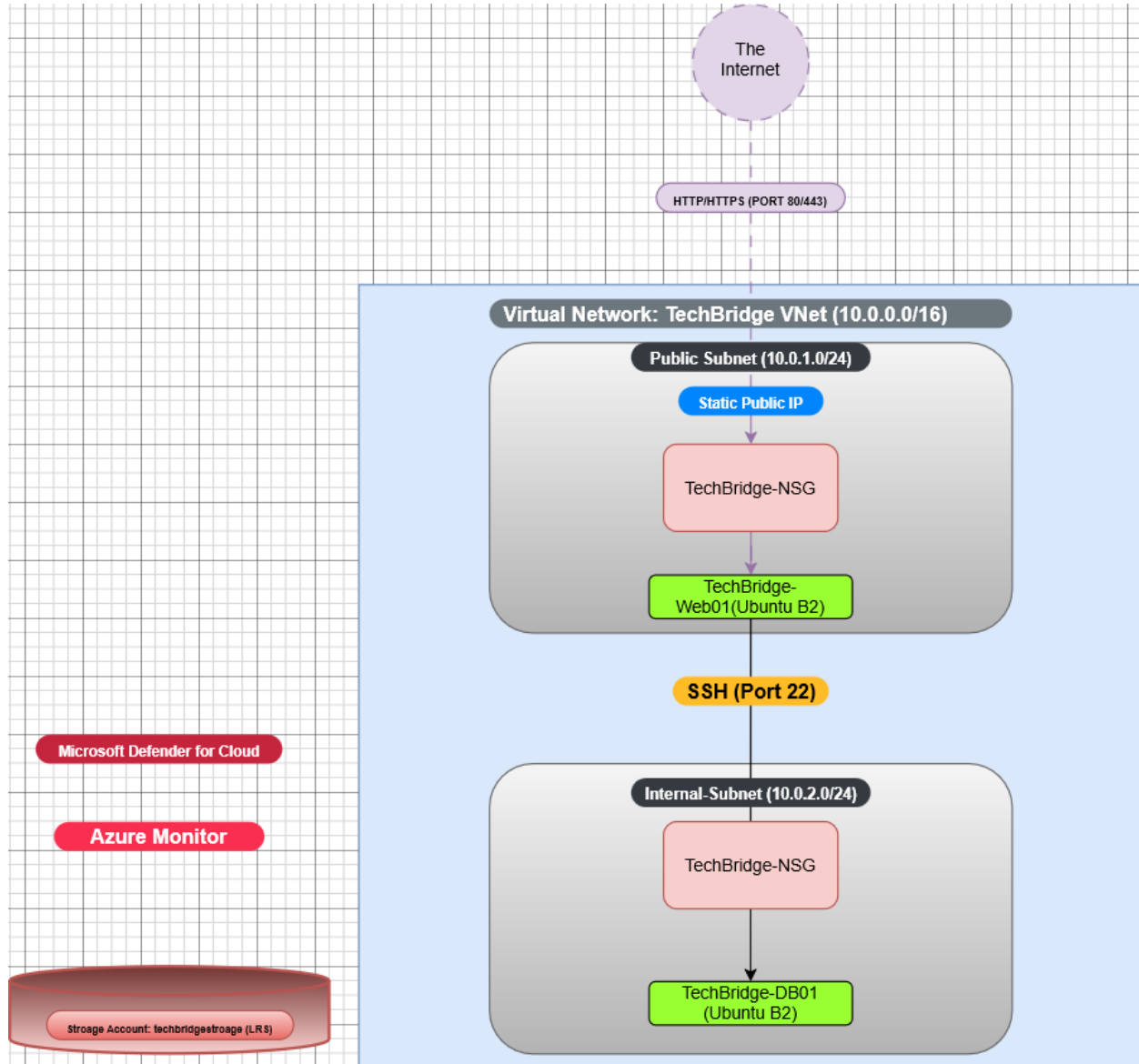


Week 9 Practical: Final Documentation

1 – 7 May 2026 | Total Marks: 14

Student Name: **Aiden Horn** Student No: **20240429**

Task 1: Create an Architecture Diagram (4 Marks)



[TechBridge.drawio](https://techbridge.drawio)

Task 2: Document Your Full Azure Environment (8 Marks)

Section 1: Resources and Configurations

Azure Service	Purpose / Why you need it	Week it is used
Resource Group	A logical container that groups all the capstone resources, makes it easier to find resources and understand billing better.	Week 1
Virtual Network (VNet)	A secure and isolated network for private communications between the virtual machines and cloud services.	Week 2
Virtual Machine (VM – B2s)	The compute infrastructure to host our web application or test server while cutting operating costs.	Week 3
Microsoft Entra ID	Used to manage user identities and use multi factor authentication and also we will use it to control access polices for students and staff.	Week 4
Network Security Group (NSG)	Acts like the virtual firewall to filter inbound and outbound traffic and using the strict allow and deny routing rules.	Week 2
Azure Storage Account	Provides a scalable object storage for the application data, the virtual machine diagnostics and the automated backups.	Week 5
Microsoft Defender for Cloud	Always monitoring the security posture of the infrastructure and this provides an actionable threat protection alert.	Week 7
Azure Monitor	This collects all the vital telemetry data to keep track of the performance and health of the network resources.	Week 8

Azure Cost Management	Tracks all cloud spending against the preset budgets which prevent billing over charges.	Week 1
-----------------------	--	--------

Resource Name	Type	Region	SKU / Config
TechBridge-RG	Resource Group	South Africa North	N/A
TechBridge-VNet	Virtual Network	South Africa North	10.0.0.0/16
TechBridge-Web01	Virtual Machine	South Africa North	Standard_B2s
TechBridge-DB01	Virtual Machine	South Africa North	Standard_B2s
techbridgestorage	Storage Account	South Africa North	Standard LRS

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for various services like Application Templates, All resources, Resource groups, App Services, Function App, Azure SQL Database, Azure Cosmos DB, Virtual machines, Load balancers, Storage accounts, Virtual networks, Microsoft Entra ID, Monitor, Advisor, Microsoft Defender for Cloud, Cost Management + Billing, and Help + support. The main content area is titled 'Resource Manager | Resource groups' and displays a table of resource groups. The table has columns for Name, Subscription, and Location. Three resource groups are listed: defaultresourcegroup-jnb, NetworkWatcherRG, and TechBridge-RG, all associated with 'Azure subscription 1' and located in 'South Africa North'. The interface includes search bars, filters, and a 'Group by' dropdown set to 'none'.

Name	Subscription	Location
defaultresourcegroup-jnb	Azure subscription 1	South Africa North
NetworkWatcherRG	Azure subscription 1	South Africa North
TechBridge-RG	Azure subscription 1	South Africa North

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Group by none

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Subscription (move) : Azure subscription 1 Deployments : 1 Failed, 8 Succeeded

Subscription ID : fe708a59-55df-4107-809f-c6f23483af31 Location : South Africa North

Tags (edit) : Add tags

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name ↑	Type	Location
MSVM-southafricanorth-techbridge-web01	Data collection rule	South Africa North
TechBridge-DB01	Virtual machine	South Africa North
TechBridge-DB01-ip	Public IP address	South Africa North
techbridge-db01312	Network Interface	South Africa North
TechBridge-DB01_OsDisk_1_99d24c4b2d5a4e968acc4	Disk	South Africa North
TechBridge-NSG	Network security group	South Africa North
TechBridge-VNet	Virtual network	South Africa North
TechBridge-VNet-bastion	Bastion	South Africa North
TechBridge-Web01	Virtual machine	South Africa North
TechBridge-Web01-CPU	Metric alert rule	Global
TechBridge-Web01-ip	Public IP address	South Africa North
techbridge-web01899	Network Interface	South Africa North
TechBridge-Web01_OsDisk_1_0d4ab01b9f6c42798448	Disk	South Africa North
techbridgestorage01	Storage account	South Africa North

Showing 1 - 14 of 14. Display count: 80...

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure Upgrade

Search resources, services, and docs (Ctrl+D)

Copy

projen, quiver1@icloud...
DEFAULT DIRECTORY: C:\WINDOWS

Home > Resource Manager > Resource groups > TechBridge-RG

TechBridge-VNet Virtual network

Search

Move Delete Refresh Give feedback

Overview

Essentials

Resource group (move) : TechBridge-RG

Location (move) : South Africa North

Subscription (move) : Azure subscription 1

Subscription ID : fe708a59-55df-4107-809f-c6f23483af31

Address space : 10.0.0.0/16

Advertised gateway prefixes : -

Subnets : 3 subnets

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : 2cc48d08-4403-4e6b-b3b5-d5497685cc82

Tags (edit) : Add tags

Topology Properties Capabilities Recommendations Tutorials

DDoS protection
Configure additional protection from distributed denial of service attacks.
Not configured

Azure Firewall
Protect your network with a stateful L3-L7 firewall.
Not configured

Peering
Seamlessly connect two or more virtual networks.
Not configured

Microsoft Defender for Cloud
Strengthen the security posture of your environment.

Private endpoints
Privately access Azure services without sending traffic across internet.
Not configured

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure | Upgrade | Search resources, services, and docs (0+)

Home > TechBridge-NSG

TechBridge-NSG

Network security group

How do I create an alert to track firewall metric failures

Search

Move Delete Refresh Give feedback

Overview

Essentials

Resource group: [TechBridge-NSG](#)

Location: South Africa North

Subscription: [Azure subscription 1](#)

Subscription ID: fe70ba59-55df-4107-809f-c9c3483af31

Tags: [Add tags](#)

Custom security rules: 4 inbound, 0 outbound

Associated with: 2 subnets, 0 network interfaces

JSON View

Filter by name

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
100	Allow-Web-Inbound	80,443	Any	Any	Any	Allow
200	SSH	22	TCP	197.184.87.135	Any	Allow
300	RDP	3389	TCP	197.184.87.135	Any	Allow
4096	Deny-All-Inbound	8080	Any	Any	Any	Deny
65000	AllowVnetInbound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInbound	Any	Any	AzureLoadBalancer	VirtualNetwork	Allow
65500	DenyAllInbound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutbound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutbound	Any	Any	Any	Internet	Allow
65500	DenyAllOutbound	Any	Any	Any	Any	Deny

Add or remove favorites by pressing ctrl + click on star

Microsoft Azure | Upgrade | Search resources, services, and docs (0+)

Home > TechBridge-NSG

techbridgestorage01

Storage account

Upload Open in Explorer Delete Move Refresh Open in mobile CU / PS Feedback

Overview

Essentials

Resource group: [TechBridge-NSG](#)

Location: southafricanorth

Subscription: [Azure subscription 1](#)

Subscription ID: fe70ba59-55df-4107-809f-c9c3483af31

Disk state: Available

Tags: [Add tags](#)

Performance: Standard

Replication: Locally-redundant storage (LRS)

Account kind: StorageV2 (general purpose v2)

Provisioning state: Succeeded

Created: 5/14/2026, 10:15:01 PM

JSON View

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace: Disabled

Default access tier: Hot

Blob anonymous access: Disabled

Blob soft delete: Enabled (7 days)

Container soft delete: Enabled (7 days)

Versioning: Disabled

Change feed: Disabled

NFS v3: Disabled

Allow cross-tenant replication: Disabled

Storage tasks assignments: None

File service

Identity-based access: Not configured

Default share-level permissions: Disabled

Soft delete: Enabled (7 days)

Require Encryption in Transit for SMB (Security): Enabled

Queue service

CMS support: Disabled

Security

Require secure transfer for REST API operations: Enabled

Storage account key access: Enabled

Minimum TLS version: Version 1.2

Infrastructure encryption: Disabled

Networking

Public network access: Enabled

Public network access scope: Enable from all networks

Private endpoint connections: 0

Network routing: Microsoft network routing

Endpoint type: Standard

Add or remove favorites by pressing ctrl + click on star

Microsoft Azure | Upgrade | Search resources, services, and docs (G+V) | Copilot | gggiles_quiver1@icloud... | Remote Desktop | Microsoft Edge

Home > TechBridge-RG

TechBridge-DB01

Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale | Open in mobile | Feedback | CU / PS

Overview

Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Monitor | Resource visualizer | Connect | Networking | Settings | Availability + scale | Security | Backup + disaster recovery | Operations | Monitoring | Automation | Help

Essentials

Resource group	techbridge-rg	Operating system	Linux (ubuntu:24.04)
Status	Running	Size	Standard_B2ats_v2
Location	South Africa North	Primary NIC public IP	20.164.210.124 1 associated public IP
Subscription	Azure subscription 1	Virtual network/subnet	TechBridge-VNet/Internal-Subnet
Subscription ID	fc708a59-55d8-4107-809f-c6923483a511	DNS name	Not configured
		Health state	1 -
		Time created	14/05/2026, 18:37 UTC

Tags (edit) | Add tags

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Virtual machine

Computer name	TechBridge-DB01
Operating system	Linux (ubuntu:24.04)
VM generation	V2
Agent status	Ready
Agent version	2.15.1.3
Hibernation	Disabled
Host group	-
Host	-
Proximity placement group	-
Colocation status	N/A
Capacity reservation group	-
Disk controller type	SCSI

Azure Spot

Azure Spot	-
Azure Spot eviction policy	-

Networking

Public IP address	20.164.210.124 (Network interface techbridge-db01312)
Public IP address (IPv6)	-
Private IP address	10.0.2.4
Private IP address (IPv6)	-
Virtual network/subnet	TechBridge-VNet/Internal-Subnet
DNS name	Configure

Size

Size	Standard_B2ats_v2
------	-------------------

Source image details

Source image publisher	canonical
Source image offer	ubuntu-24.04-ifs
Source image plan	server

JSON View

Add or remove favorites by pressing Ctrl+V on the left

Microsoft Azure | Upgrade | Search resources, services, and docs (G+V) | Copilot | gggiles_quiver1@icloud... | Remote Desktop | Microsoft Edge

Home > TechBridge-RG

TechBridge-Web01

Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale | Open in mobile | Feedback | CU / PS

Overview

Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Monitor | Resource visualizer | Connect | Networking | Settings | Availability + scale | Security | Backup + disaster recovery | Operations | Monitoring | Automation | Help

Essentials

Resource group	techbridge-rg	Operating system	Linux (ubuntu:24.04)
Status	Running	Size	Standard_B2ats_v2 (2 vcpus, 1 GB memory)
Location	South Africa North	Primary NIC public IP	-
Subscription	Azure subscription 1	Virtual network/subnet	-
Subscription ID	fc708a59-55d8-4107-809f-c6923483a511	DNS name	-
		Health state	1 -
		Time created	14/05/2026, 18:35 UTC

Tags (edit) | Add tags

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Virtual machine

Computer name	TechBridge-Web01
Operating system	Linux (ubuntu:24.04)
VM generation	V2
VM architecture	x64
Agent status	Ready
Agent version	2.15.1.3
Hibernation	Disabled
Host group	-
Host	-
Proximity placement group	-
Colocation status	N/A
Capacity reservation group	-
Disk controller type	SCSI

Azure Spot

Azure Spot	-
------------	---

Networking

Public IP address	-
Public IP address (IPv6)	-
Private IP address	-
Private IP address (IPv6)	-
Virtual network/subnet	-
DNS name	-

Size

Size	Standard_B2ats_v2
vCPUs	2
RAM	1 GB
Threads per core	2

Source image details

Source image publisher	canonical
Source image offer	ubuntu-24.04-ifs

Section 2: Security Configuration

Network Security Group Rules

Inbound Security Rules				
Rule Name	Port	Protocol	Source	Allow/Deny
Allow-Web-Inbound	80, 443	Any	Any	Allow
SSH	22	TCP	My IP 197.184.87.135	Allow
RDP	3389	TCP	My IP 197.184.87.135	Allow

My security groups rules follow the principle of least privilege

RBAC using Entra ID users and groups

Home > TechBridge-RG

TechBridge-NSG

Network security group

How do I create an alert to track firewall metric failures

×

Search

MoveDeleteRefreshGive feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Essentials

Custom security rules : 4 inbound, 0 outbound

Associated with : 2 subnets, 0 network interfaces

JSON View

Filter by name

Port == allProtocol == allSource == allDestination == allAction == all

Priority	Name	Port	Protocol	Source	Destination	Action	
Inbound Security Rules							
100	Allow-Web-Inbound	80,443	Any	Any	Any	Allow	
200	SSH	22	TCP	197.184.87.135	Any	Allow	
300	RDP	3389	TCP	197.184.87.135	Any	Allow	
4096	Deny-All-Inbound	8080	Any	Any	Any	Deny	
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow	
65500	DenyAllInBound	Any	Any	Any	Any	Deny	
Outbound Security Rules							
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow	
65500	DenyAllOutBound	Any	Any	Any	Any	Deny	

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

giggles_quiver1k@idou...

DEFAULT DIRECTORY (GIGGLESQ...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Home

Users

Default Directory

+ New user

Edit

Delete

Download users (Preview)

Bulk operations

All users

Azure Active Directory is now Microsoft Entra ID.

Search users

Add filter

7 users found

☐	Display name ↑	User principal name ↑	User type	Is Agent	
☐	AH Aiden Horn	giggles_quiver1k_i...	Member	No	↑
☐	CW Craig Wentworth	craigwentworth@...	Member	No	↑
☐	DG Dave Green	davegreen@giggl...	Member	No	↑
☐	JS Jason slot	jasonslot@giggles...	Member	No	↑
☐	JB Jim Brown	jimbrown@giggle...	Member	No	↑
☐	MV Mike Van Rensburg	mikevanresburg@...	Member	No	↑
☐	TV taylin van niekerk	taylinvanniekerk@...	Member	No	↑

Add or remove favorites by pressing Ctrl+Shift+F



TechBridge-RG | Access control (IAM)

☆ ...

+

Add

↓

Download role assignments

≡

Edit columns

↻

Refresh

🗑

Delete

🗨

Feedback

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription

9 / 4000

Privileged

5

View assignments

Type : All

Role : All

Scope : All scopes

Group by : Role

All (9)

Job function roles (4)

Privileged administrator roles (5)

☐

Name

↑↓

Type

↑↓

Role

↑↓

Scope

↑↓

Condition

↑↓

Owner (1)

☐

Aiden Horn

2d2850fd-4d53-41dc-...

User

Owner

Subscription (Inherited)

None

Contributor (4)

☐

Craig Wentworth

ebc1d0ab-2644-4185-...

User

Contributor

This resource

None

☐

Dave Green

9ce6d193-b2a4-47a0-...

User

Contributor

This resource

None

☐

Jim Brown

1517e0af-9589-4d74-...

User

Contributor

This resource

None

☐

Mike Van Rensburg

d55b4d57-4069-4939-...

User

Contributor

This resource

None

Reader (2)

☐

Jason slot

dac34239-a94b-4077-...

User

Reader

This resource

None

☐

Operations and Finance

0cc90ec2-6688-4254-...

Group

Reader

This resource

None

Virtual Machine Contributor (2)

☐

Craig Wentworth

ebc1d0ab-2644-4185-...

User

Virtual Machine Contributor

This resource

None

☐

taylin van niekerk

7a69fcef-85b4-4e4b-...

User

Virtual Machine Contributor

This resource

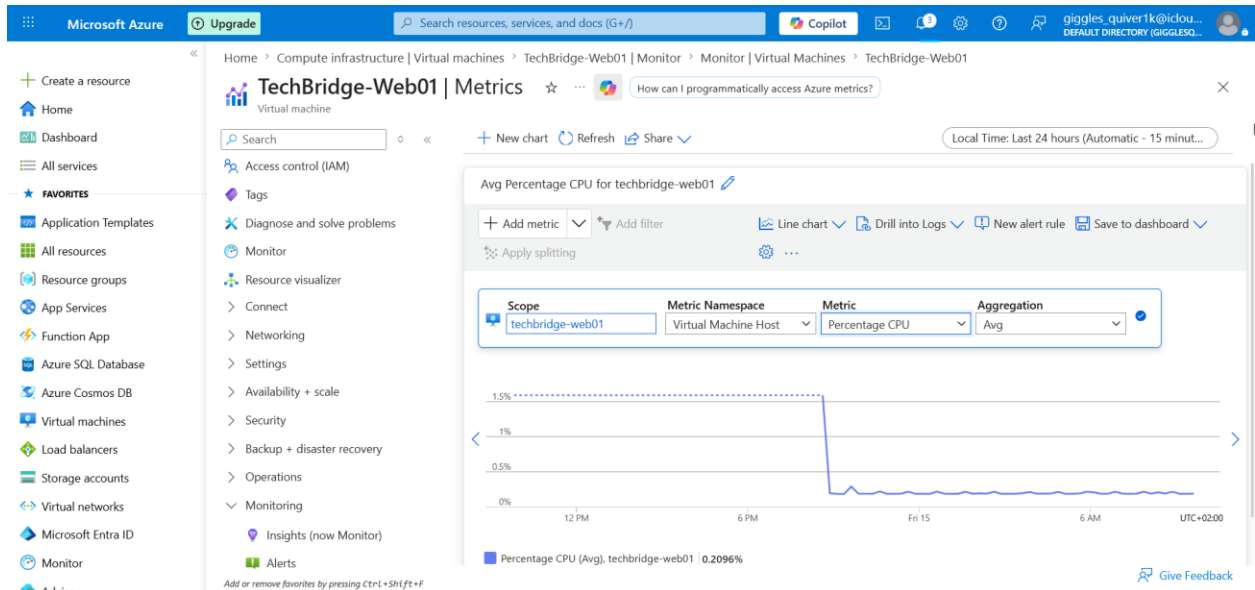
None

Showing 1 - 9 of 9 results.

Add or remove favorites by pressing Ctrl+Shift+F

Section 3: Monitoring and Alerting

CPU percentage is being monitored and will alert via email when it reaches 80% or higher And budget alert is set to stay under \$100 and alert once it reaches 80% and also a forecasted total.



Create budget ...

Budget
expiration date and the amount.

*Name

Capstone-Budget

✓

*Reset period ⓘ

Monthly

▼

*Creation date ⓘ

2026

▼

May

▼

1

*Expiration date ⓘ

2028

▼

April

▼

30

▼

Budget Amount

Give your budget amount threshold

Amount *

100

✓

VIEW OF MONTHLY COST DATA

Create budget ...

Budget

✓ Create a budget 2 Set alerts

Configure alert conditions and send email notifications based on your spend.

* Alert conditions

Type	% of budget	Amount	
Forecasted cost	80	80	
Actual cost	100	100	
Select type	Enter %	-	

* Alert recipients (email)

Alert recipients (email)

example@email.com

Previous

Create

Microsoft Azure

Upgrade

Search resources, services, and docs (G + /)

Copilot

giggles.quiver1k@iclou...
DEFAULT DIRECTORY (GIGGLESC...)

Create a resource

Home

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All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Home > Cost Management + Billing | Cost Management > Cost Management: Aiden Horn

Cost Management: Aiden Horn | Budgets

Show my Credit Balances

What are my top 5 services with

Billing account

Search

+ Add

Refresh

Feedback

Help

Scope: Aiden Horn

Search by name

All periods

Name	Scope	Reset period	Creation date	Expiration date	Budget	Forecasted	E
Capstone-Budget	11643017-c53f-5267...	Monthly	01/05/2026	30/04/2028	US\$100.00	ⓘ	\$

Budget was created

Budget Capstone-Budget was successfully created.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+I)

Copilot

giggles.quiver1k@iclou...
DEFAULT DIRECTORY (GIGGLESC...

Create a resource

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Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Home > Compute infrastructure | Virtual machines > TechBridge-Web01 | Monitor > Monitor | Virtual Machines > TechBridge-Web01 | Alerts

Alert rules

+ Create ☆ Set up recommended alerts | Columns Refresh Export to CSV Open query Delete Enable Disable

New: Let Observability Agent reason over your alerts to automatically correlate alert noise for faster investigations. Request preview access

Search Subscription: Azure subscription 1 Target resource type: all Add tag filter More (4) No grouping

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ TechBridge-Web01-CPU	Percentage CPU > 80	3 - Informational	techbridge-web01	Virtual machine	Metrics	Enabled

Showing 1 - 1 of 1 results. Give feedback

Microsoft Azure

Upgrade

Search resources, services, and docs (G+I)

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

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Resource groups

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Azure SQL Database

Azure Cosmos DB

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Load balancers

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Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Home > Compute infrastructure | Virtual machines > TechBridge-Web01 | Monitor > Monitor | Virtual Machines > TechBridge-Web01 | Alerts

Create an alert rule

Scope Condition Actions Details Tags Review + create

Product details

Metric alert rule

1 Condition

Terms of use | Privacy statement

Total pricing

0.10 USD/month

Pricing

Scope

Scope level

Resource

Subscription

Azure subscription 1 > techbridge-rg > techbridge-web01

Condition

Signal name

Operator

Aggregation type

Threshold value

Lookback period

Check every

Percentage CPU

Greater than

Average

80

5 minutes

1 minute

Details

Project details

Subscription

Resource group

Region

Azure subscription 1

TechBridge-RG

global

Alert rule details

Alert rule name

Alert rule description

Severity

Enable upon creation

Automatically resolve alerts

TechBridge-Web01-CPU

3 - Informational

☒

☒

Create

Previous